The O'Neil method, letter by letter

Above: RESEARCH-WORKER.md §3.2 (the steps), which points here for the method behind each letter, its rules, thresholds and traps.

Talks to:

- the research worker, which runs these computations and writes each as a dated fact;
- the desk's card and market reader, which show the readings;
- qp's backtest, whose CANSLIM gate reads the same facts as of a day.

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1. What it is

William O'Neil's CANSLIM, reconstructed from free, official data. CANSLIM is a way of finding leading growth stocks by seven marks: **C**urrent quarterly earnings, **A**nnual earnings, **N**ew (a product, management, industry conditions, or a price high out of a base), **S**upply and demand, **L**eader or laggard, **I**nstitutional sponsorship, and **M**arket direction.

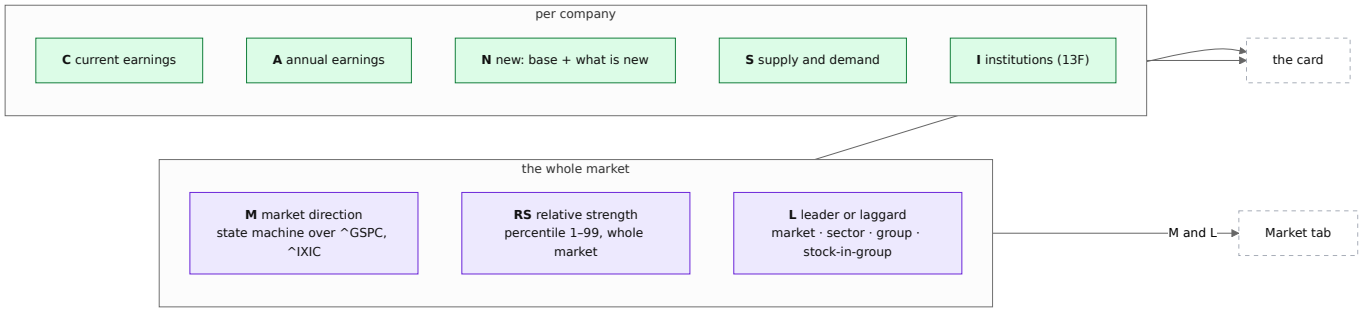
Three rules govern the whole reconstruction:

- **Reconstructed, and named as such.** Every reading is built from public data and called what it is (`rs` , never "IBD RS"); the numbers are ours, their provenance stated, because membership and sources differ from IBD's.
- **A label, never a filter.** No tool's candidates change because of a letter; the readings label a card and fill the Market tab (ARCHITECTURE §7).
- **O'Neil's own numbers,** as named constants, printed beside any signal that fires, so a reading can be checked against its definition.

What it is not is as important as what it is: most of CANSLIM's ideas share a name with a common indicator that measures something else (§3).

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2. The diagram



The whole-market letters (M, RS, L) exist for every US stock the moment it is carded; the per-company letters (C, A, N, S, I) are computed for the tickers the tools find (RESEARCH-WORKER.md §3.2).

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3. The traps the method must avoid

Most CANSLIM concepts have a plain-English name that means something else in common use. Implementing the common reading gives a number that looks right and measures a different thing.

Concept	What it is	What it is mistaken for
RS Rating	a percentile 1–99 of 12-month performance against **every other stock**	RSI, which compares a stock to its own range
C	current quarter EPS vs **the same quarter a year ago**	sequential quarters (reintroduces seasonality)
A	annual EPS growth over 3–5 years, plus ROE: durability	a longer C
Distribution day	one session: index down ≥ 0.2 % on **higher volume**	any down day
Distribution count	days **live** in a rolling 25-session window, with expiry	a running total
A/D Rating	a letter A–E from 13 weeks of price and volume	Chaikin's A/D index, an unbounded cumulative line
Follow-through day	a confirmation on **day 4+** of a rally attempt: a big gain on higher volume	any strong up day
N: new high	price emerging from a **base** into new ground, on volume	any 52-week high
Group rank	rank of the stock's **industry group**	the sector (far too coarse)
M	a **rule-based state machine** over two indexes	a sentiment read or an MA cross
RS line	the plotted ratio stock ÷ index; "new high **before** price" is the tell	the RS Rating
U/D volume ratio	up-day volume ÷ down-day volume over 50 days	the A/D rating
Base stage	which base, counted **from the market bottom**	any consolidation
Group rank vs Group RS	**one fact**: rank 63 of 197 *is* the 68th percentile *is* B+	three measurements that could disagree
RS within the group	"RS 1 of 13": the stock ranked against its own group only	the RS Rating
N: the new thing	**three** things: product, management, **better industry conditions**	two things, and a news feed

Each row has a fixture in the worker's tests (RESEARCH-WORKER.md).

4. The letters

4.1 M: the market model

A state machine over the **S&P 500 (^GSPC)** and the **Nasdaq Composite (^IXIC)**, daily bars. Three published states (confirmed uptrend, uptrend under pressure, market in correction) and one internal (rally attempt).

Rule	Value	Provenance
Distribution day	close ≤ prior close × (1 − 0.002) **and** volume > prior volume, while in an uptrend	IBD's −0.2 % floor on O'Neil's "closes lower on higher volume"
Stalling day	volume heavy, gain under ~0.2 %, close in the lower half of the range: counts as distribution, flagged apart	IBD
Live window	25 sessions	IBD; the tight end of O'Neil's "5–6 weeks"
Removal	after 25 sessions, **or** when the index **trades** 5 % above that day's close	IBD
Under pressure	5 live distribution days	IBD
Correction	6+ live, or a decisive break of the rally low	IBD
The bottom	the **lowest close** of the correction (not an intraday low)	the 2010 workshop
Rally day 1	the session **after** that lowest close; a new lower close moves the bottom and restarts the count	the workshop
Follow-through	day **4 or later**, gain **≥ 1.7 %**, volume **> the prior session** (that is the whole test)	O'Neil's last figure (1 % → 1.5 % → 1.7 %)
Late follow-through	days 8–11 late, beyond 11 very late: **flagged, never refused**	refusing creates a state the market can never leave
Status	counted **per index**, published as the **worse** of the two	"one index is enough"

Lesson from the first live run. An extra test (volume above the 50-day average) blocked the Nasdaq's August 2026 follow-through; the model said "correction" through a confirmed rally. A rule stricter than the published one fails to the wrong answer, in the direction of telling you not to buy. The 50-day comparison is reported beside a follow-through, never decides it. And the caveat is always shown: about one follow-through in four fails — it is necessary for a bottom, not sufficient.

4.2 RS: the relative strength rating

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RS = 0.4·(P0/P63) + 0.2·(P0/P126) + 0.2·(P0/P189) + 0.2·(P0/P252)
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63 / 126 / 189 / 252 trading days = 3 / 6 / 9 / 12 months, the latest quarter weighted double; then **percentile-ranked 1–99 against every US stock**. Verified against IBD's published formula. Two choices make it match: the universe is the whole market (Polygon grouped daily,

every US ticker each session), and prices are **split-adjusted** — a 2-for-1 split would otherwise score a leader at −50 %.

4.3 L: groups and the four levels of strength

Level	The number	How	Answers
Market	the M state + the index RS line	§4.1; index ÷ S&P 500	buy at all?
Sector	rank of a coarse sector	aggregate performance	which half of the market
Group	rank X of N	**median** member RS, membership by SIC	the leading group? (O'Neil: ~37 % of a stock's move; sector ~12 %)
Stock in group	"RS 1 of 13"	the stock's RS ranked inside its group	leader or laggard

Membership is the SIC code from EDGAR for every filer (TradingView labels only where SIC has none — one writer, F-21). **Thin groups** (fewer than 6 ranked members) roll up one rung (4-digit industry → 3-digit group → 2-digit major group), and are dropped if still thin. Groups are ranked by median RS; **percentile and letter** (A+ ≥ 94 … E) are derived from the rank, never stored as separate facts — rank 63 of 197 *is* the 68th percentile *is* B+ (one fact, three prints). **Rotation**: rank now against rank three months ago. **The divisor is stored with every rank** (IBD once went from 197 groups to 145), so the card prints `63 of 197`, never a bare rank. The honest limit: our group rank is ours, because membership differs; the rank within the group and the member count are exact.

4.4 C and A: earnings, from EDGAR

From the SEC's company-facts API (official, free, a descriptive User-Agent; CIK ↔ ticker from `company_tickers.json`).

- **C** (current quarter): eight quarters of EPS, the year-ago EPS and %Chg, sales, year-ago sales and %Chg, after-tax margin; **accelerating** (each of the last 2–3 quarters' %Chg larger than the one before); **beat +25 %**, X of 8.
- **A** (annual): fiscal years of EPS and %Chg (with the year's price high and low), the year **rebuilt from four quarters** to check against the filing, **3-year growth**, **stability** (the wobble around trend, 0–99, low is good), and **ROE** against O'Neil's **17 %** floor.

The five EDGAR traps, each of which produces a normal-looking wrong number:

#	Trap	Rule
1	C is year over year, never sequential	compare to the quarter one year back (within 45 days)
2	Q4 is usually not filed as a quarter	Q4 = FY − Q1 − Q2 − Q3, **only when all three are present**; left out when EPS and net income disagree in sign
3	the same period is filed more than once	the most recently **filed** value wins
4	a % from a negative base is not a number	`n/a (loss a year ago)` ; capped at ±999 %
5	revenue lives under four tags	read all four, take one per period

Filings land weeks after quarter end, so **every block carries its own as-of date** (RESEARCH-WORKER.md §3.3).

4.5 N: the base, and what is new

The weekly base (weekly, not a display choice: a base is 7–65 weeks, and daily bars drown every shape test in noise): weeks, depth, waves down (the decline typically comes in three), **accumulation weeks** (heavy volume, no price progress), the **handle** (a controlled 2–8-week drift in the **upper half** — a handle that gives back half the cup is the cup failing), lip and low with their weeks, the **pivot**, % away, X of 6 checks. Needs 7 weekly bars.

The RS line tell: the RS line at a new high **while price is still in its base** fires; both at new highs does not (that is arithmetic); the line off its high does not. **Divergence:** the index made a new low in the window, the stock did not, and the stock is above where it sat on the index's worst close — distinct from "the index has not made a new low".

What is new (the N pipeline, designed, not built — §8): three things, not two. New product/service, new management, **better industry conditions** (the group's rank rising), harvested from EDGAR 8-K items, openFDA and USASpending, scored by the price reaction and whether the RS line held. News APIs corroborate, never trigger.

4.6 S: supply and demand

Shares outstanding (from EDGAR, its basis when not the cover figure, with an as-of); **share-count change over about a year**, always from the same tag (buying back / diluting / flat, or the reason there is no rate); float (TradingView) and float as a % of shares; short interest and days to cover (from the card's short-interest facts); **U/D volume ratio** over 50 sessions; and **A/D**, a bounded 13-week score (−1 … +1, normalised by the window's volume so it compares between stocks), banded to a letter that is **ours** (IBD does not publish its boundaries), with the raw number beside it.

4.7 I: institutional sponsorship, from Form 13F

The SEC's quarterly 13F datasets. Counted as distinct **managers** by CIK, unioned across every CUSIP of the issuer (stock, converts, options are one company) and across a 13F-HR and its amendment (one manager). Per quarter: the holder count, **of N** managers who filed at all, and the **share of filers**. **Direction is read from the share**, with a 3 % relative dead band: `rising` (up over the year and still up), `peaked` (up over the year, newest quarter down), `flat`, `falling`, `turning up`; only `rising` and `turning up` are coloured bullish, with the raw holder change

printed beside, uncoloured. Only managers over $100M file, 45 days after quarter end: the lag is stated on screen.

4.8 On each card: the market reflected

Four per-card readings, each a different number on every card:

1. **Behaviour on the live distribution days:** on how many the stock beat the index close-to-close, and the average difference, with the dates behind a tap; it says so when there are none.
2. **Sessions since the follow-through,** banded early (< 25), established (25–150), late (> 150).
3. **Group rotation:** rank now vs three months ago, ▲ into / ▼ out of.
4. **The exposure implication,** in O'Neil's words.

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5. Contracts

The research worker is the only writer; each letter is one fact family, keyed by `market` or by ticker (RESEARCH-WORKER.md §4):

Letter	Fact family
M	`oneil_m` (with `bars_universe` behind RS and `group_ranking` behind L)
RS	`rs_rating`, `rs_line`
L	`oneil_l`, `group_ranking`, `sic`
C, A	`oneil_c`, `oneil_a`
N	`oneil_n_base`, `oneil_n_events`
S	`oneil_s`
I	`oneil_i`
ratings	`oneil_ratings` (RS, group, A/D, volume vs 50-day; later EPS, SMR, Composite)

Everything explains itself. Each fact family ships with a definition card: what it means, how it is calculated, **what it is not** (the trap in §3), its source and its as-of. The card and the Market tab show it on a tap.

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6. When it fails

Event	What happens
a source is slow or limited (EDGAR: ten requests a second)	the step works within its budget and carries the rest to the next night; the fact keeps its old date and the card says "last run DATE" (RESEARCH-WORKER.md §5)
no filings for a company	the fact's status is `none`, not a guessed 0
a % from a negative base	`n/a (loss a year ago)`, not a wild number
a follow-through is late (days 8+)	it is flagged late, never refused
a group is thin	it rolls up a rung, or is dropped — never ranked on too few members

Honest gaps everywhere: a missing reading says it is missing and why, and is never shown as 0, "no", or a bad sign.

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7. Built on

The old desk: legacy/qp/CANSLIM.md, 1,653 lines, the deepest document of the old desk — the traps, the exact rules and their provenance, the card and panel specification, and the build phases. The code the worker reuses: `oneil.py`, `relstrength.py`, `groups.py`, `sic.py`, `edgar.py`, `f13.py`, `base.py`, `ratings.py`, `canslim.py`.

What failed there (the method's faults; the worker's own faults are in RESEARCH-WORKER.md §6):

Fault	Prevented by
a rule stricter than O'Neil's (a 50-day volume test) blocked a real follow-through and held the model in "correction"	only the published test decides; the extra is reported, never deciding
the industry map had two writers (SIC and TradingView) and flipped every morning (F-21)	one writer, SIC first, TradingView only where SIC has none
three separate fields for group rank, percentile and letter invited a card showing them disagree	one fact, three prints, derived from the rank
CANSLIM rebuilt "as of" a day separately in the backtest	dated facts the gate reads by `as_of`

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8. Status

Built, in shadow (RESEARCH-WORKER.md §7): M (the whole market model), groups and the four levels, A/D, U/D, the RS line tell and divergence, the EDGAR C and A tables, shares outstanding and the weekly base, and 13F sponsorship.

To build, each behind its own versioned definition:

#	What
1	the **index RS line**, and **group new highs / lows** (cheap, from data already fetched)
2	the **sector rank** from SIC major groups, kept distinct from the group
3	**EPS rating** and **SMR** from EDGAR, then **Composite** once both exist (a composite without EPS and SMR would be a partial one wearing a complete name)
4	the **float estimate** (shares outstanding minus insider and 5 %-holder positions), labelled an estimate; funds / banks / management %
5	the **N pipeline** (§4.5): harvest → classify → react → persist → score → one line, with discarded counts printed as proof it is a filter, not a feed

`rs_line`, `oneil_s`, `oneil_n_events` and `sic` as dated facts, and the ratings beyond RS, group and A/D, are the other **to build** items the parent names.